

Normal Operations Checklist

Date: Thursday, 13 February 2025

Time completed 17:11 UTC

Warning: Invalid values present

VTOL ENOC v5 MK9 SPARTA

Start	
A1) Mission name	MK9
A2) GCS Version	3.4.1.1053
A3) APS Version	4.2.3+85f18c94+0+official
Local Time	02/13/2025 07:48:59
A5) Gimbal extended and external fan turned on (if applicable)	True
A6) Start Screen Recorder	True

Location	
A7) Latitude	39.35 °
A8) Longitude	-9.32 °
A9) Altitude	-3.61 ft AMSL
A10) Home location altitude	38.93

Tracker ON	
A11) Operational Mode	Normal

Comms	
A12) Link Quality - BackupRadio	100 %
A12) Link Quality - PrimaryRadio	100 %
A13) Local RSSI - PrimaryRadio	dBm
A14) Remote RSSI - PrimaryRadio	dBm

Vehicle	
A15) Flight Mode	Manual
A16) Power Source	External
A17) Voltage	0.1 V
A18) Current	0.4 A
A19) EAS*	0.75 kn
A20) GS*	0.23 kn
A21) Vehicle Altitude*	3 ft ARL

Parameters	
A22) All Parameters downloaded*	True
A30) ARSPD_PRIMARY	0
A31) ARSPD_AUTOCAL	0
A32) ARSPD2_AUTOCAL	0
A33) ARSPD3_AUTOCAL	
A34) ALT_HOLD_RTL	-1
A35) ABAG_ENABLED	1
A36) Q_ENABLE	1
A37) FTLMODE5	20
A38) FTLMODE6	20

Flight Control	
A23) Engine Ignition*	On
A24) Use Compass	1
A25) Use Airspeed	1
A26) Initial Fuel Level Updated	7.3 L
A27) FPV Camera On	True
A28) Lights On	True

Transponder	
A29) Turn on and set up transponder (if applicable)	True

Airspeed Sensors	
A37) Airspeed 1	4.52 kn
A38) Airspeed 2	4.75 kn
A39) Airspeed 3	0 kn

GPS	
A40) GPS Sats - 2	29
A40) GPS Sats - 1	12
A41) GPS Lock* - 2	Dgps
A41) GPS Lock* - 1	Fix3D
A42) HDOP* - 2	0.51
A42) HDOP* - 1	0.54

Environment Sensors	
A43) Outside Pressure	1020.37 hPa
A43) Outside Pressure	1020 hPa
A43) Outside Pressure	1020.68 hPa
A44) Mag X - 3	0 mT
A44) Mag X - 2	452 mT
A44) Mag X - 1	-79 mT
A45) Mag Y - 3	0 mT
A45) Mag Y - 2	322 mT
A45) Mag Y - 1	-192 mT
A46) Mag Z - 3	0 mT
A46) Mag Z - 2	127 mT
A46) Mag Z - 1	272 mT
A47) Vehicle Properly Aligned*	True

IMU Sensors ACC	
A48) Attitude Sensors Coherent*	True
A49) Acc X - 3	18 mg
A49) Acc X - 2	35 mg
A49) Acc X - 1	36 mg
A50) IMU ACC1, ACC2 and ACC3 X Max. difference +/-50*	True
A51) Acc Y - 3	-1 mg
A51) Acc Y - 2	13 mg
A51) Acc Y - 1	4 mg
A52) IMU ACC1, ACC2 and ACC3 Y Max. difference +/-50*	True
A53) Acc Z - 3	-993 mg
A53) Acc Z - 2	-981 mg
A53) Acc Z - 1	-981 mg

IMU Sensors Gyros	
A54) Gyro X - 3	-1 mRad s
A54) Gyro X - 2	0 mRad s
A54) Gyro X - 1	-1 mRad s
A55) Gyro Y - 3	2 mRad s
A55) Gyro Y - 2	-1 mRad s
A55) Gyro Y - 1	0 mRad s
A56) Gyro Z - 3	0 mRad s
A56) Gyro Z - 2	0 mRad s
A56) Gyro Z - 1	-1 mRad s

Mission Planning

Areas	
B1) NOTAM and Areas Loaded	True

Home	
B2) Set Home Location	True
B3) Set Home Altitude	True
B4) Check Aircraft Altitude	True

ROUTE PLANNING - Takeoff	
B5) - WP1 1 - Position for VTOL launch 2 - Set Altitude to 150 ft ARL 3 - Set Waypoint Type: VTOL Take-off 4 - Set Pre-Waypoint Action to: NO ACTION 5 - Set Take-Off Yaw: 0 (default)	True
B6) - WP2 1 - Set Position: set accordingly 250m aligned with the wind from VTOL take-off 2 - Set Altitude: same as above 3 - Set Waypoint Type: Standard 4 - Set Pre-Waypoint Action to: NO ACTION	True

B7) - WP3	True
1 - Set Position: set accordingly after leaving WP2	
2 - Loiter outside of WP2 Acceptance radius	
3 - Set Altitude: minimum 250ft	
4 - Set Waypoint Type: Loiter Altitude	
5 - Set correct loiter direction: CW / CCW	
6 - Set Pre-Waypoint Action to: NO ACTION	

ROUTE PLANNING - Safety Loiter	
B8) - WP4	True
1 - Set Position: set accordingly	
2 - Set Altitude: 350ft	
3 - Set Waypoint Type: Loiter	
4 - Set Loiter Radius: 250m	
5 - Set Pre-Waypoint Action to: NO ACTION	

ROUTE PLANNING - Landing

B9) - WP5 Prepare Landing 1 - Set Position: set accordingly 2 - Set Waypoint Type: Loiter Altitude 3 - Set Altitude: 200ft 4 - Set Pre-Waypoint Action to: PREPARE LANDING	True
B10) - WP6 Aligment 1 - Set Position: set accordingly, wind aligned 2 - Set Aligment WP 350m away from the Landing WP 3 - Set Altitude: 150 ft ARL 4 - Set Waypoint Type: Standard 5 - Set Pre-Waypoint Action to: NO ACTION	True

B11) - WP7 Land	True
1 - Set Position: set accordingly, Place to Land	
2 - Place the Landing point 350m away from the Alignment WP	
3 - Set Altitude: 0 ft ARL	
4 - Set Waypoint Type: VTOL landing	
5 - Set Approach alt 0, Landing Yaw, 0- NO ACTION	

Finalise	
B12) Upload Route and Re-Check*	True
B13) Rename and Save Routes (if applicable)	True

RallyPoints	
B14) Set or Confirm Rallypoints*	True

Pre-Engine Start Checks

Flight Mode - Manual	
C1) Flight mode*	Manual

RC	
C2) Battery Voltage*	8 V
C3) Confirm prop covers are on*	True
C4) Arm VTOL motors	True
C5) Check correct movement of control surfaces/throttle*	True
C6) Throttle Percentage in FlyByWireA	100
C7) Flight mode changes in FGCS	True
C8) Disarm VTOL motors*	True

Base Test	
C9) Altitude approx. 0 ft ARL on ground*	True
C10) Altitude approx. 6 ft ARL on ground*	True

Stabilisation	
C11) Flight Mode - FlyByWire B	FlyByWireB

C12) 1 - Roll Right	True
2 - Roll Left	
C13) 1 - Pitch Up	True
2 - Pitch Down	
C14) 1 - Rudder Right	True
2 - Rudder Left	
C15) Airspeed Test more than 47kn (24m/s)	54.51 kn

Reset Mode to Manual	
C16) Flight Mode	Manual

LIDAR check	
C17) LIDAR reading properly*	0.28 ft

VTOL MOTORS - Batteries	
C18) 1 - Aircraft on the Stand and attached	True
2 - Connect VTOL batteries	
3 - Confirm beeps and slight rotation on motors	
C19) Confirm battery is secured and battery cover is closed	True
C20) VTOL battery 1 charged*	49.5
C21) VTOL battery 2 charged*	49.5
C22) VTOL ESC Current - FL *	0.04
C23) VTOL ESC Current - RL *	0.04
C24) VTOL ESC Current - FR*	0.04
C25) VTOL ESC Current - RR*	0.04

VTOL Test	
C26) 1 - Confirm Area Cleared	True
2 - Set Throttle Idle	
3 - VTOL prop covers OFF	
C27) Change Mode to VTOL STABILISE on GCS	VtolStabilise
C28) ARM VTOL motors	True

C29) Motors:	True
1 - Started	
2 - React according SP inputs	
3 - Direction of rotation correct - check sticker on the arms	
C30) DISARM VTOL*	True

VTOL Test - End	
C31) Flight Mode	Manual

Departure Position	
C32) VTOL prop covers ON*	True
C33) Move aircraft to Departure place	True

Payload	
C34) Start Payload Checklist	True

Lights	
C35) Lights On	True

Engine Start Checks

Start	
D1) Flight Mode (Manual)	Manual
D2) 1 - Confirm aircraft secured	True
2 - Set Throttle Idle	
3 - Arm VTOL Motors	
4 - Request Engine Start	
D3) Engine Start Time	02/13/2025 08:10:05

Warmup	
D4) Engine warmed up* - Primary	100.1 °C

MAX RPM	
D5) Throttle Percentage in MANUAL	100
D6) Engine Max. RPM*	7727

D7) Fuel Flow	1.92 L/h
D8) Onboard Generator*	charging
D9) IAT Sensor	20.6

MAX RPM - BACK to IDLE	
D10) CHT Sensor <200°C - Primary	200 °C
D11) Payload Retracted*	True
D12) 1 - Artificial Horizon	True
2 - Cowling System OK	
3 - Smoke and Noise OK*	

DISARM	
D13) DISARM VTOL*	True

Idle	
D14) Engine Idle RPM*	3107

Departure Checks

Gimbal	
E1) Payload Checklist Complete*	True
E2) 1 - Gimbal Protect	True
2 - Gimbal retracted	

Completing ENOC	
E3) ENOC Finished	True
E4) Operator responsible	Pedro Luis
E5) Publish the checklist!	True